

How many parameters do we have to introduce to form general solution of a linear system, and how do we tell which variables to be assigned as parameters?

Answer:

The answers lie in the structure of the REF or RREF of the augmented matrix.

Every column of the REF/RREF corresponds to a variable of the system, except for the last (augmented) column which corresponds to the constant terms of the system.

Columns that **contain a pivot** (leading entry) are known as **pivot columns**; while

Columns that **do not contain a pivot** are known as **non-pivot columns**.

The number of parameters you must introduce to form a general solution of a linear system is given by the simple formula:

Number of Parameters = (Total Number of variables of the system) - (Number of Pivot Columns in the REF/RREF)

To follow the convention, the variables that should be assigned parameters are all the variables that correspond to non-pivot columns of the REF/RREF. However, it is also possible to replace such variables with variables that correspond to the pivot columns that are closest to the non-pivot columns on their left. To illustrate with an example, suppose C_1, C_2, C_3, C_4, C_5 are the five columns in the REF/RREF that correspond to the variables x_1, x_2, x_3, x_4, x_5 of the system, and C_1, C_3, C_4 are the pivot columns. By convention, we will assign x_2 and x_5 as parameters since C_2 and C_5 are the non-pivot columns. Alternatively, we can also assign x_4 as parameter in place of x_5 , since C_4 is the nearest pivot column on the left of C_5 ; and assign x_1 as parameter in place of x_2 , since C_1 is the nearest pivot column on the left of C_2 .

Scenario 1

Consider a system with 4 variables and 3 equations.
$$\begin{cases} x_1 + 2x_2 + 0x_3 + x_4 = 7 \\ x_1 + 2x_2 + x_3 + 4x_4 = 12 \\ 2x_1 + 4x_2 + 0x_3 + 3x_4 = 16 \end{cases}$$

The augmented matrix is:
$$\begin{bmatrix} 1 & 2 & 0 & 1 & 7 \\ 1 & 2 & 1 & 4 & 12 \\ 2 & 4 & 0 & 3 & 16 \end{bmatrix}$$

After performing Gauss-Jordan elimination, we arrive at the

following RREF:
$$\begin{bmatrix} \mathbf{1} & 2 & 0 & 0 & 5 \\ 0 & 0 & \mathbf{1} & 0 & -2 \\ 0 & 0 & 0 & \mathbf{1} & 2 \end{bmatrix}$$

The pivot columns are those columns with leading entries 1 in each row, as shown in bold.

- Total number of variables = 4 (x_1, x_2, x_3, x_4)
- Number of pivot columns = 3 (columns 1, 3, 4)

Therefore, the number of parameters needed is: 4 (total) - 3 (pivots) = 1. We will need one parameter.

By convention, we assign a parameter to the variable x_2 corresponding to the non-pivot column 2.

Let $x_2 = s$, where s is a parameter, representing any real number.

Now, we write out the equations from the RREF matrix:

1. $x_1 + 2x_2 + 0x_3 + 0x_4 = 5$
2. $0x_1 + 0x_2 + x_3 + 0x_4 = -2$
3. $0x_1 + 0x_2 + 0x_3 + x_4 = 2$

We solve for the variables x_1, x_3, x_4 in terms of the parameter s :

- From eq (1): $x_1 + 2(s) = 5 \Rightarrow x_1 = 5 - 2s$
- From eq (2): $x_3 = -2$
- From eq (3): $x_4 = 2$

The general solution of the system is given by

- $x_1 = 5 - 2s$
- $x_2 = s$
- $x_3 = -2$
- $x_4 = 2$

Alternatively, we can also assign x_1 as a parameter in place of x_2 , since column 1 is the nearest pivot column on the left of column 2.

Let $x_1 = s$, where s is a parameter.

Now, we write out the equations from the RREF matrix:

1. $x_1 + 2x_2 + 0x_3 + 0x_4 = 5$
2. $0x_1 + 0x_2 + x_3 + 0x_4 = -2$
3. $0x_1 + 0x_2 + 0x_3 + x_4 = 2$

We solve for the variables x_2, x_3, x_4 in terms of the parameter s :

- From eq (1): $s + 2x_2 = 5 \Rightarrow x_2 = 5 - \frac{1}{2}s$
- From eq (2): $x_3 = -2$
- From eq (3): $x_4 = 2$

The general solution of the system is given by

- $x_1 = s$
- $x_2 = 5 - \frac{1}{2}s$
- $x_3 = -2$
- $x_4 = 2$

(Note that general solution of a linear system is an expression to represent all possible solutions of the system. Such an expression is not unique.)

Scenario 2

Consider a system with 5 variables and 2 equations.
$$\begin{cases} x_1 + 2x_2 - x_3 + 0x_4 + 3x_5 = 4 \\ 2x_1 + 4x_2 - x_3 + x_4 + 8x_5 = 11 \end{cases}$$

The augmented matrix is:
$$\left[\begin{array}{ccccc|c} 1 & 2 & -1 & 0 & 3 & 4 \\ 2 & 4 & -1 & 1 & 8 & 11 \end{array} \right]$$

After performing Gaussian elimination, we arrive at the

following REF: $\begin{bmatrix} \mathbf{1} & 2 & -1 & 0 & 3 & 4 \\ 0 & 0 & \mathbf{1} & 1 & 2 & 3 \end{bmatrix}$

The pivot columns are those columns with leading entries 1 in each row, as shown in bold.

- Total number of variables = 5 (x_1, x_2, x_3, x_4, x_5)
- Number of pivot columns = 2 (columns 1, 3)

Therefore, the number of parameters needed is: 5 (total) - 2 (pivots) = 3. We will need three parameters.

By convention, we assign parameters to the variables x_2, x_4, x_5 corresponding to the non-pivot columns 2, 4, 5.

Let $x_2 = s, x_4 = t, x_5 = r$ where s, t, r are parameters.

Now, we write out the equations from the RREF matrix:

1. $x_1 + 2x_2 - x_3 + 0x_4 + 3x_5 = 4$
2. $0x_1 + 0x_2 + x_3 + x_4 + 2x_5 = 3$

We solve for the variables x_1, x_3 in terms of the parameters s, t, r :

- From eq (2): $x_3 + t + 2(r) = 3 \Rightarrow x_3 = 3 - t - 2r$
- From eq (1): $x_1 + 2(s) - (3 - t - 2r) + 3(r) = 4 \Rightarrow x_1 = 7 - 2s - t - 5r$

The general solution of the system is given by

- $x_1 = 7 - 2s - t - 5r$
- $x_2 = s$
- $x_3 = 3 - t - 2r$
- $x_4 = t$
- $x_5 = r$

Alternatively, we can also assign x_1 as a parameter in place of x_2 , since column 1 is the nearest pivot column on the left of column 2; and assign x_3 as a parameter in place of either x_4 or x_5 since column 3 is the nearest pivot column on the left of column 4 and 5. To illustrate, let $x_1 = s, x_3 = t, x_4 = r$ where s, t, r are parameters.

Now, we write out the equations from the RREF matrix:

1. $x_1 + 2x_2 - x_3 + 0x_4 + 3x_5 = 4$
2. $0x_1 + 0x_2 + x_3 + x_4 + 2x_5 = 3$

We solve for the variables x_2, x_5 in terms of the parameters s, t, r :

- From eq (2): $t + r + 2(x_5) = 3 \Rightarrow x_5 = \frac{1}{2}(3 - t - r)$
- From eq (1): $s + 2x_2 - t + r + 2(\frac{1}{2}(3 - t - r)) = 4 \Rightarrow x_2 = \frac{1}{2}(1 - s - 2t)$

The general solution of the system is given by

- $x_1 = s$
- $x_2 = \frac{1}{2}(1 - s - 2t)$
- $x_3 = t$
- $x_4 = r$
- $x_5 = \frac{1}{2}(3 - t - r)$